
ISyE 6740 - Spring 2021

Project Proposal

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Project Title: Project to Predict 2024 United States Election using Naive Bayes

Problem Statement

The United States presidential election on November 4th, 2024, is a significant topic for Americans, with a high number of undecided voters and considerable unpredictability in the outcome. This project aims to analyze the 2020 presidential election results, demographic data, economic indicators, and polling results from various sources to model a prediction of who has the highest probability of winning the 2024 election. Additionally, it will investigate which features provide the most valuable insights for predicting the outcome.

This project aims to raise voter awareness and support informed voting decisions by helping readers understand the likely outcomes of the election. The results can foster debate and discussion around the latent variables analyzed in this report, ultimately assisting voters in identifying the factors that most influence a presidential election.

Data Source

The project will incorporate various data sources with the objective of balancing the features that contribute to predicting the 2024 U.S. election results. Detailed links are provided at the end of the document.

Demographic information, such as age, gender, income levels, and education for both 2020 and 2024, will be gathered from the United States Census Bureau via their API. Information on the 2020 election results, as well as campaign finance data for 2020 and 2024, will be sourced from the Federal Election Commission and the National Archives websites. Polling information from 2020 and 2024 will be collected from the Pew Research Center. Economic indicator data will be obtained from the U.S. Bureau of Economic Analysis (BEA). Social media sentiment analysis will be conducted using the X

(formerly Twitter) API. Lastly, news articles related to the election will be scraped using website APIs and BeautifulSoup.

Methodology

Naive Bayes and logistic regression will be the primary classifiers used to predict the election outcome. We will utilize the ‘GaussianNB’ and ‘LogisticRegression’ modules to train the models. Naive Bayes is considered a classifier with a low misclassification rate, making it potentially the best method for this task. We will compare its results with a logistic regression classifier to validate the approach. The features from different sources will be collected, and we will explore various dimensionality reduction techniques, such as PCA for Naive Bayes and Lasso or Ridge regression for logistic regression. PCA will enable us to perform feature selection while retaining the maximum variability in the data. We chose PCA because it is known to help reduce the risk of overfitting. Additionally, we will analyze collinearity among the features and use PCA to minimize multicollinearity, ensuring that the model remains robust and interpretable.

Sentiment analysis will be performed on social media platforms, such as X (formerly Twitter), using NLTK by creating a corpus of the most relevant topics over time. The results from the sentiment analysis will then be merged with the other variables in the dataset.

We will experiment with simulation statistical techniques such as Monte Carlo simulation to examine the role of randomness in our analysis and evaluate its influence on potential outcomes across multiple scenarios. We plan on incorporating this methodology as we considered it will be useful in an election since the outcome will more likely vary depending on randomness. By generating a large number of random samples and running repeated simulations, we can approximate the distribution of outcomes and assess how different inputs or assumptions impact the final results. This approach will help us in highlighting the probability of different outcomes and offering insights into the robustness and sensitivity of the model in the face of unpredictable factors.

The Python packages we intend to use include ‘scikit-learn’, ‘statsmodels’, ‘xgboost’, ‘pandas’, ‘matplotlib’, ‘seaborn’, ‘tweepy’, ‘requests’, ‘BeautifulSoup’ and ‘nltk’.

Evaluation and Final Results

The Naive Bayes model will be compared with the logistic regression model and evaluated using a confusion matrix, accuracy metrics, precision, recall, and F1-score. The coefficients of the features will be analyzed and evaluated to interpret the importance of each feature. We will use decision boundaries after PCA to visually assess how well the models separate the classes. In addition to this, we plan to

visualize the ROC curve and AUC to assess the trade-off between sensitivity and specificity, indicating which model performs better. Cross-validation will be used to measure performance across different datasets.

Links to Data Sources:

Data source from the Federal Election Commission (FEC): <https://www.fec.gov/introduction-campaign-finance/election-results-and-voting-information/federal-elections-2020/>

Data source from the National Archives: <https://www.archives.gov/electoral-college/2020>

Polling information from the Pew Research Center: <https://www.pewresearch.org/dataset/2024-national-public-opinion-reference-survey-npors/>